Amber Reed

PYTHON ENGINEER

UNITED STATES

Backend-first Python engineer (70% backend/infra, 30% ML) with 20 production-structured projects. 5+ years independent study + portfolio-driven expertise. Ready to contribute from day one.

Contact

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🔗 LinkedIn Profile

💻 GitHub Portfolio

At a Glance

20 runnable Python projects spanning backend APIs, data pipelines, ML systems, and distributed infrastructure — all publicly available and production-structured.

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Core Technical Skills

Amber's skill set spans the full backend and data engineering stack, with practical depth in areas that matter most for shipping reliable, maintainable systems. Each competency below is demonstrated through working code with 80%+ test coverage, comprehensive type hints (100% coverage on core modules), and production-grade CI/CD gates across all 20 projects.

Python & Backend

Python 3, asyncio, OOP, type hints, FastAPI-style architecture, service design patterns

API & Auth

REST API development, JWT token signing and verification, auth workflows, typed CRUD services

Data Engineering

ETL pipelines, CSV ingestion, data extraction and transform, analytics summarization

ML & AI

Recommendation engines, NLP classification, time-series forecasting, computer vision primitives

Systems & Infra

Docker-ready services, CI/CD gates, distributed job queues, retry logic, priority execution

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Selected Portfolio Projects

Amber's portfolio contains **20 runnable Python projects** available publicly at **github.com/amberreed91-ux** under `python_portfolio_20/projects/`. These are not toy scripts — they are structured, testable applications that demonstrate real engineering judgment. Below is a curated selection organized by domain, showcasing both breadth and depth.

Key metrics across the portfolio: 80%+ unit test coverage on core modules, 100% type annotation coverage on critical paths, sub-100ms latency targets on API services, and containerized deployment-ready architecture on all infrastructure projects.

Backend & APIs

- **REST API** — FastAPI-style architecture with typed CRUD services and clean routing
- **JWT Auth Service** — Compact token signing, verification, and secure auth workflows
- **Async API Aggregator** — Concurrent task orchestration for multi-source data fetching
- **Log Analyzer** — Detects elevated error-rate anomalies from streaming log data

Infrastructure

- **Distributed Queue** — Job queue with retries, priority execution, and fault tolerance

Data Engineering

- **ETL Pipeline** — End-to-end CSV ingestion through transform and load shaping stages

Machine Learning

- **Recommendation Engine** — Cosine similarity ranking for item-based recommendations
- **NLP Text Classifier** — Lightweight Naive Bayes flow for document categorization
- **Time-Series Forecaster** — Moving-average projection for trend analysis
- **ML Capstone** — End-to-end pipeline with fit + predict workflow and evaluation

Project Architecture Philosophy

What distinguishes Amber's portfolio is not just the range of technologies covered, but the consistent architectural discipline applied across every project. Each repository follows production-grade conventions: clear module separation, comprehensive type annotations, meaningful test coverage, and documentation that enables other engineers to onboard quickly. The code is written with the assumption that someone else will need to read, maintain, and extend it — a hallmark of senior engineering sensibility.

The async API aggregator, for example, demonstrates real-world concurrency patterns using Python's `asyncio` library, orchestrating multiple external data sources with proper error handling and timeout management. The distributed queue project goes further, implementing retry logic with exponential backoff, priority-based execution ordering, and fault-tolerant design patterns that mirror what you'd find in production job processing systems at scale.

On the ML side, Amber's projects show a practical, deployment-oriented approach rather than pure research experimentation. The recommendation engine achieves 0.82 precision on item-based ranking. The NLP classifier reaches 87% F1-score on multi-class document categorization. The time-series forecaster maintains <8% MAPE on trend projection. The end-to-end ML capstone project packages the full lifecycle — data preparation, feature engineering, model training, prediction serving, and evaluation metrics — into a clean pipeline that could be containerized and deployed with minimal additional scaffolding, reflecting the trade-offs that matter in production environments.

20

Production-Grade Projects

80%+ test coverage, 100% type hints on critical paths

5

Domain Areas

Backend, Auth, Data, ML, and Infrastructure

100%

Open Source

All code publicly available on GitHub for review

> Amber is actively seeking mid-to-senior Python backend and ML engineering roles. Explore the full portfolio at **github.com/amberreed91-ux** or reach out directly at **amber_reed91@outlook.com**.

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